



Unity 20 Bluetooth Fingerprint Reader Developer's Guide

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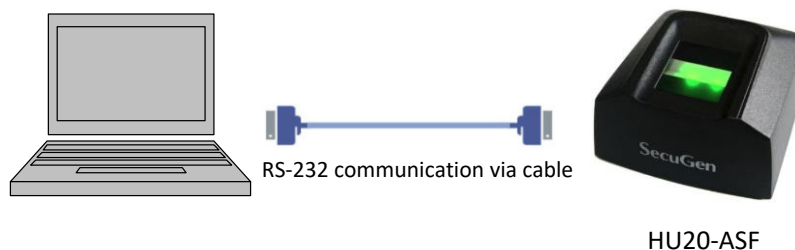
1. Product Overview

The Unity 20 Bluetooth is a fingerprint recognition system that combines a high performance optical fingerprint reader and Bluetooth module in one device. The Unity 20 Bluetooth can capture fingerprints on-the-spot at any time and transfer fingerprint templates to a host device such as a smartphone or PC via the Bluetooth interface. It features a Bluetooth interface configured to provide Serial Port Profile Services (SPP).

The Serial Port Profile defines the protocols and procedures that shall be used by devices using Bluetooth for RS232 (or similar) serial cable emulation. The scenario covered by this profile deals with legacy applications using Bluetooth as a cable replacement, through a virtual serial port.

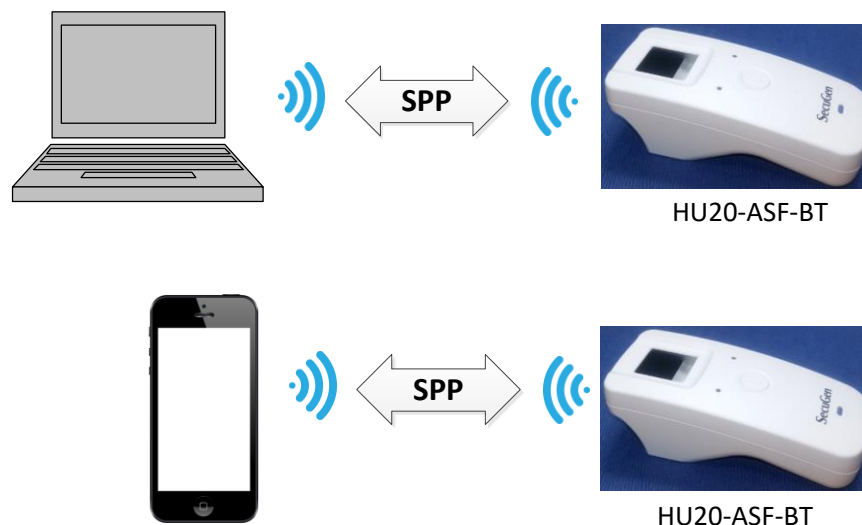
Serial communication with the HU20-ASF fingerprint sensor is performed using a cable to connect the host to the device.

Wired Serial Communication



Using the SPP profile, serial communication with the HU20-ASF-BT fingerprint sensor is performed wirelessly.

Wireless Serial Communication



Developers can build applications that connect to HU20-ASF-BT using a virtual Bluetooth COM port and send commands to the device to manage configuration, capture fingerprints, register or verify users and so-on. Please refer to the document entitled “**FMS Protocol Manual**” included with the HU20-ASF-BT DK to learn more about programming the Unity 20 Bluetooth.

2. FMS Sample Applications

The Unity 20 Bluetooth does not require any platform specific libraries for application development. Since the Bluetooth Serial Port Profile is used, the development process is fairly straightforward. A typical scenario is shown below:

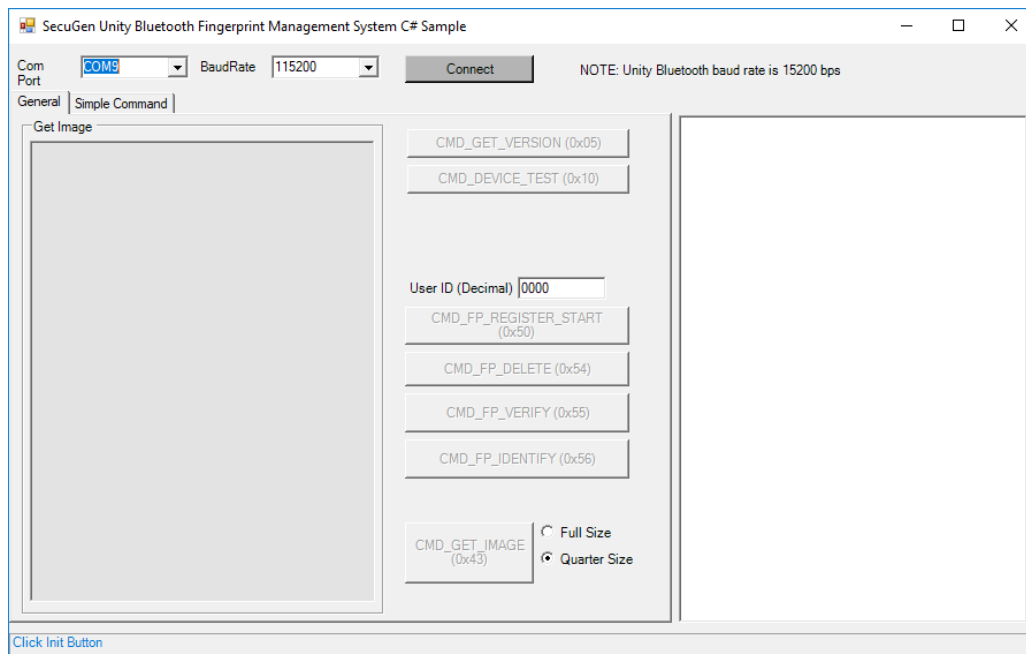
- Pair the host computer or mobile device with Unity Bluetooth
- Open a serial connection to Unity Bluetooth with port settings 115200 bps, N, 8, 1
- Send a 12-byte packet containing an FMS command to Unity Bluetooth
- Process the 12-byte response from Unity Bluetooth (Response data may be longer than 12 bytes if fingerprint image or template data is being retrieved from the device)

to build an application that is a fingerprint recognition system that combines a high performance optical fingerprint reader and Bluetooth module in one device. The Unity 20 Bluetooth can capture fingerprints on-the-spot at any time and transfer fingerprint templates to a host device such as a smartphone or PC via the Bluetooth interface. It features a Bluetooth interface configured to provide Serial Port Profile Services (SPP).

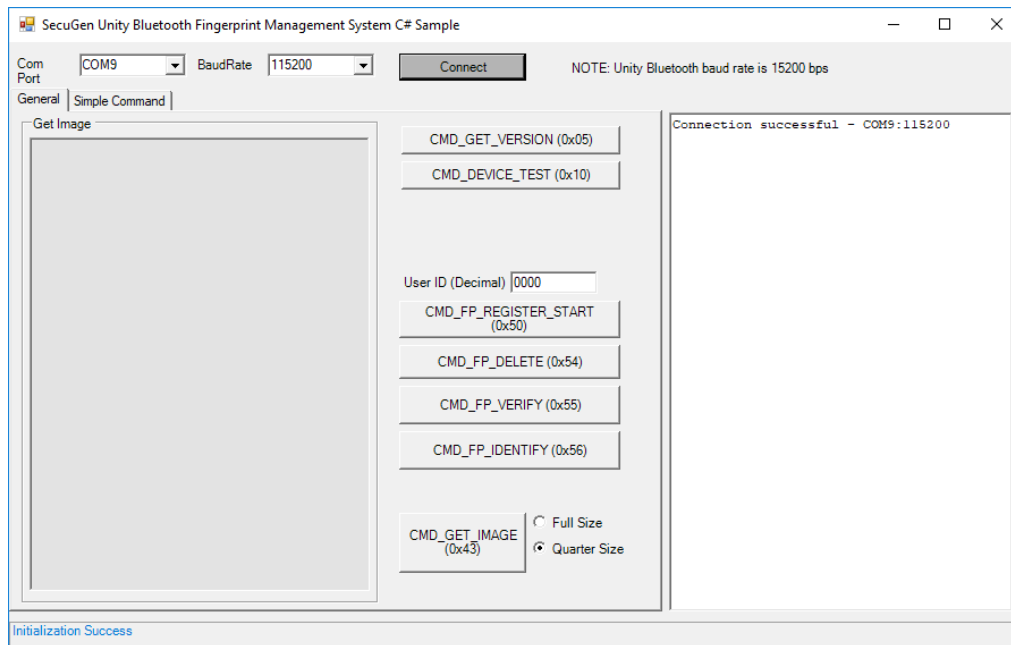
C# .NET Sample

C# sample code is included with this DK. This sample code has been built with Visual Studio 2012. Visual Studio 2015 is also supported.

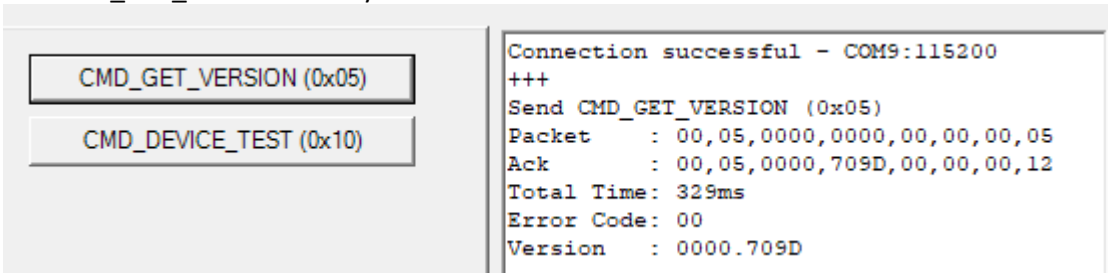
1. Open the solution in **Bluetooth_Sample_C#** folder and build it.
2. Run the application and select the COM port associated with your Unity Bluetooth device



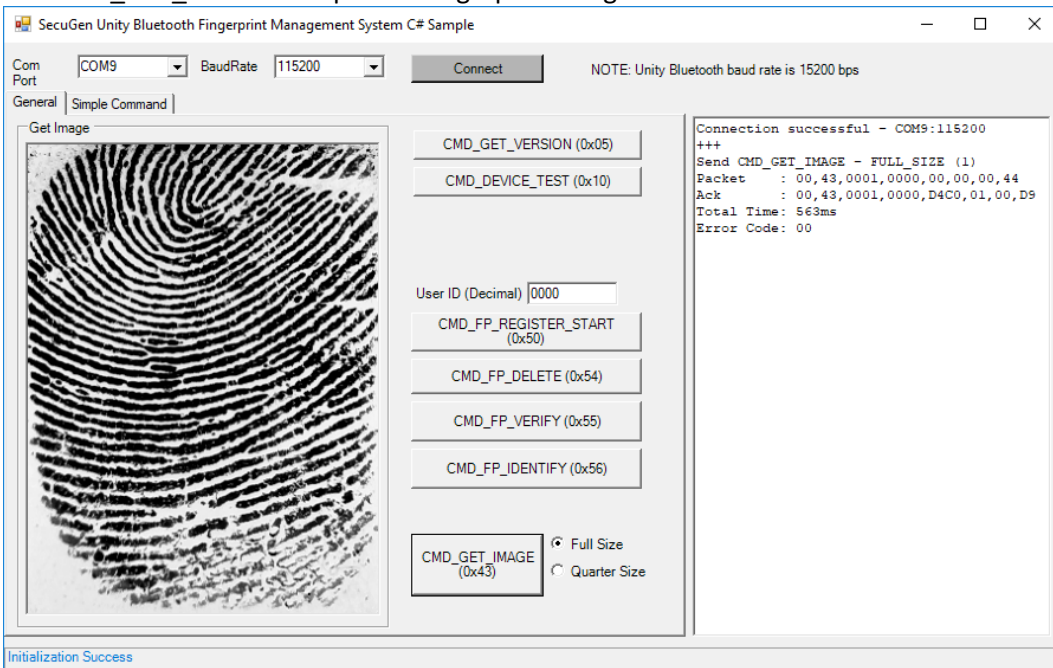
- Click Connect. Unity Bluetooth status LED should change from flashing blue to solid blue.



- Click CMD_GET_VERSION. Verify that firmware version is v0.709D or later



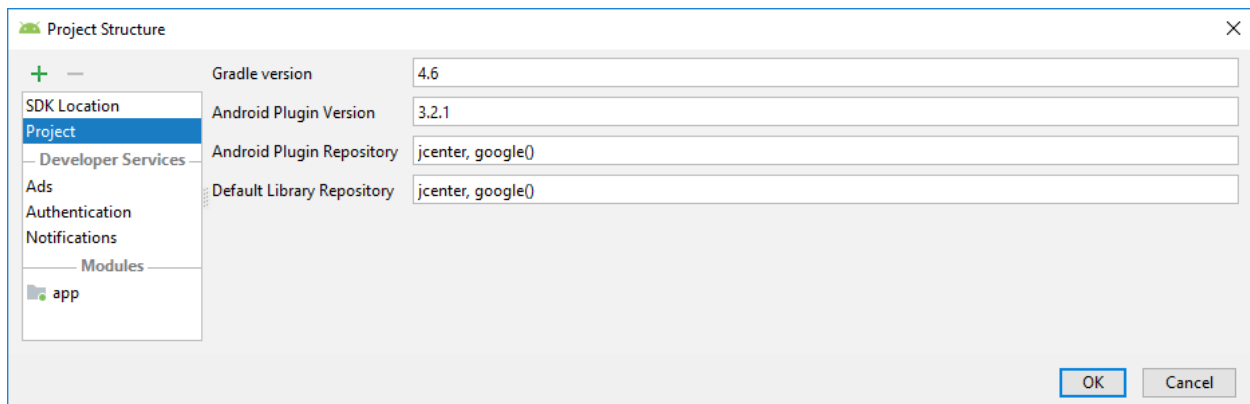
- Click CMD_GET_IMAGE to capture a fingerprint image.

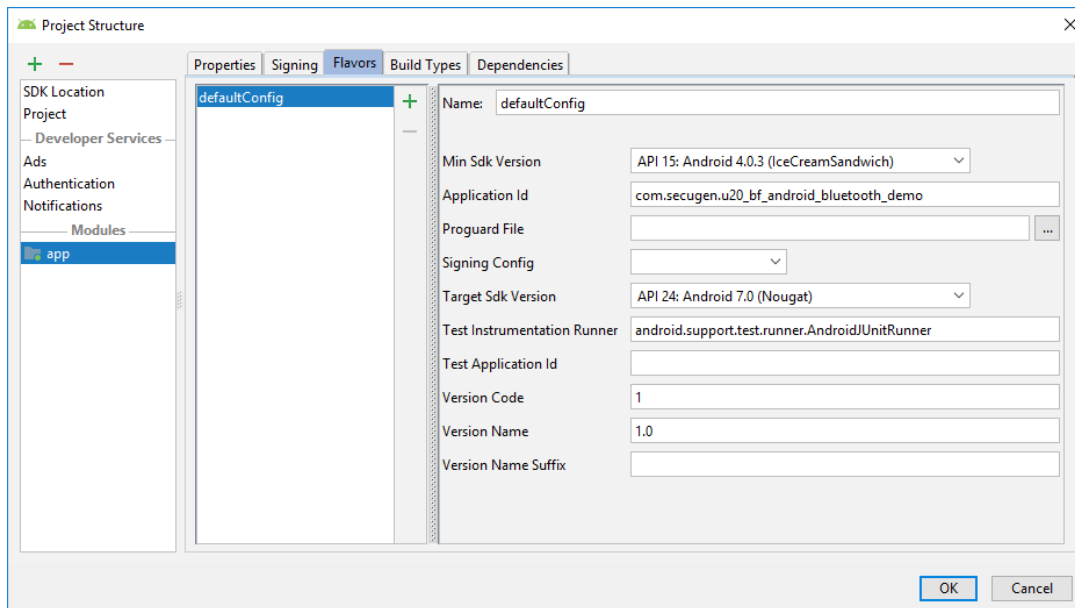
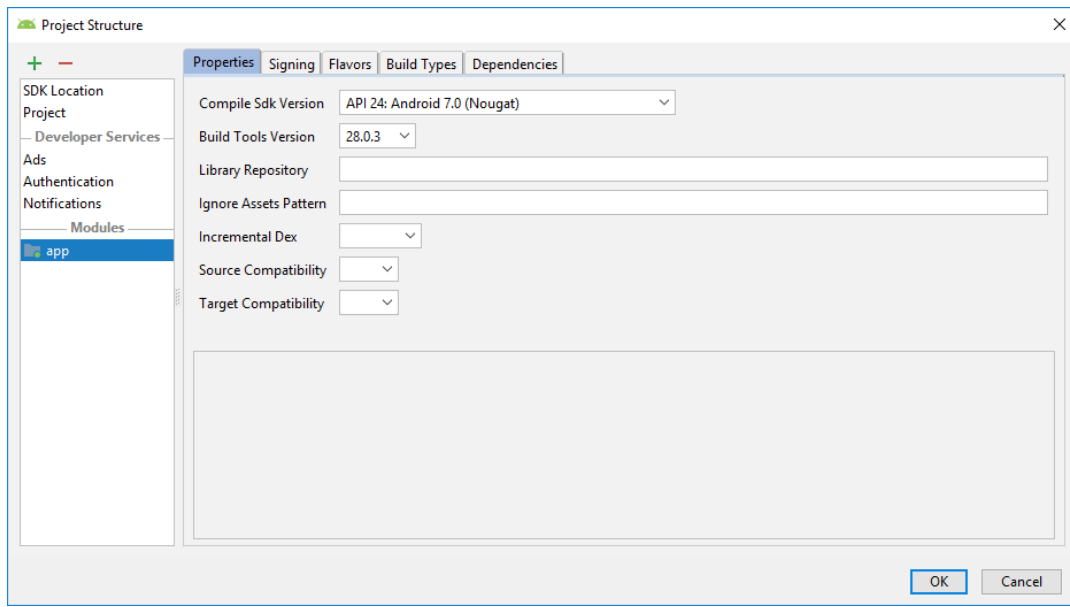


Android Studio Sample

An Android Studio sample project is included with this DK. This project will generate an application package

1. Verify that your Android Studio environment supports the tool versions mentioned below. If not, configure your environment accordingly.





2. Build the debug version of the Android sample app.
3. Locate the .apk file in the application build output directory.

U20-BF_Android_Bluetooth_Demo > app > build > outputs > apk > debug				
Name	Date modified	Type	Size	
app-debug.apk	1/4/2019 1:53 PM	APK File	1,247 KB	
output.json	1/4/2019 1:53 PM	JSON File	1 KB	

4. Install the .apk file on you Android device.

5. Launch the application and connect to Unity Bluetooth. You will then be able to register, verify and identify users on the Unity Bluetooth device.

